CS & IT ENGINEERING

COMPUTER ORGANIZATION AND ARCHITECTURE

Cache Organization

Lecture No.- 4

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Recap of Previous Lecture

Topic Cache Write

Topic Cache Mapping

Topic Direct Mapping

Topics to be Covered

Topic Direct Mapping

Topic Cache Initialization

[NAT]

1.68 ns

#Q. The memory access time is 1 nanosecond for a read operation with a hit in cache, 5 nanoseconds for a read operation with a miss in cache, 2 nanoseconds for a write operation with a hit in cache and 10 nanoseconds for a write operation with a miss in cache. Execution of a sequence of instructions involves 100 instruction fetch operations; 60 memory operand read operations and 40 memory operand write operations. The cache hit-ratio is 0.9. The average memory access time (in nanoseconds) in executing the sequence of instructions is?

	hit	miss
Read	1 ns	5 ns
write	2 ns	10 ns

$t_{avg\ read} = 0.9 * 1 + 0.1 * 5 = 1.4$ ns

$t_{avg\ write} = 0.9 * 2 + 0.1 * 10 = 2.8$ ns

Total 200 times CPU accesses mem.

Read = 160 times = 100 + 60 $\Rightarrow \frac{160}{200} = 0.8$

write = 40 times $\frac{40}{200} = 0.2$

$T_{avg} = 0.8 * 1.4\,ns + 0.2 * 2.8\,ns$

$= 1.68\,ns$

Total read time

$= 160 * 1.4\,ns$

Total write time

$= 40 * 2.8\,ns$

$avg = \frac{160 * 1.4 + 40 * 2.8}{200}$

$= 1.68\,ns$

[NAT]

#Q. Size of data sent to main memory from CPU:

1. For write hit, when a write through cache is used? $\Rightarrow$ 1 data item size
2. For write miss, when a write through cache is used? $\Rightarrow$ 1 data item size
3. For write hit, when a write back cache is used? $\Rightarrow$ nothing
4. For write miss, when a write back cache is used? $\Rightarrow$ nothing

[MCQ]

#Q. Size of data sent from main memory to cache:

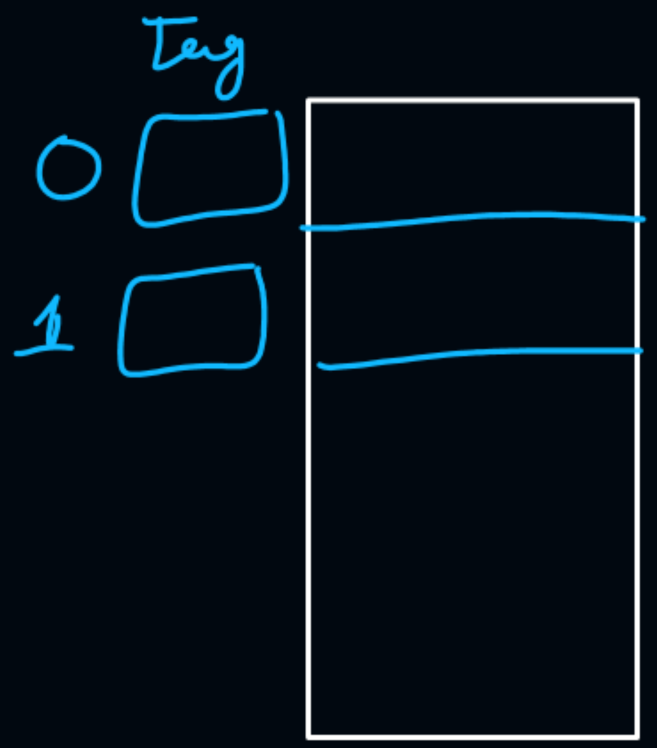
1. For write hit, when a write through cache is used? ⇒ nothing
2. For write miss, when a write through cache is used? ⇒ nothing
3. For write hit, when a write back cache is used? ⇒ nothing
4. For write miss, when a write back cache is used? ⇒ a block

Topic : Direct Mapping

Cache	0	1	2	3	4	5	6	7	8	9
Main Memory	00	01	02	03	04	05	06	07	08	09
	10	11	12	13	14	15	16	17	18	19
	20	21	22	23	24	25	26	27	28	29
	30	31	32	33	34	35	36	37	38	39
	:	:	:	:	:	:	:	:	:	:
	90	91	92	93	94	95	96	97	98	99

$$\text{cm block no.} = (\text{mm block no.}) \; \% \; (\text{no. of blocks in Cache})$$

$$\text{Tag} = \left\lfloor \frac{\text{mm block no.}}{\text{no. of blocks in cache}} \right\rfloor$$



mm block no.

Tag	cm block no.

Topic : Direct Mapping

- Blocks in cache = 4 (00-11) 2^2
- Blocks in Main memory = 8 (000-111) $\to 2^3$

Topic : Direct Mapping

mm block no.

⟵ 3-bits ⟶

Tag	cm block no.
1 bit	2 bits

mm block no = 000 ⟸ $(0)_{10}$

0	00

= 100 = $(4)_{10}$

1	00

Topic : Direct Mapping

CPU Request (MM block)	Mapping (CM block no.)	Hit /Miss	Comments
$(0)_{10} = (000)_2$	$\boxed{0 \mid 00}$ Tag, cm block no. = 0	miss	bring mm block 000 content in cache at block 00 with tag 0
$(4)_{10} = (100)_2$	$\boxed{1 \mid 00}$ Tag, cm block=0 Tag not matching	miss	bring mm block 100 content in cache at block 00 with tag 1 by replacing content of block 000 from Cache

CPU generates mm address

$\Downarrow$

extract mm block no.

$\Downarrow$

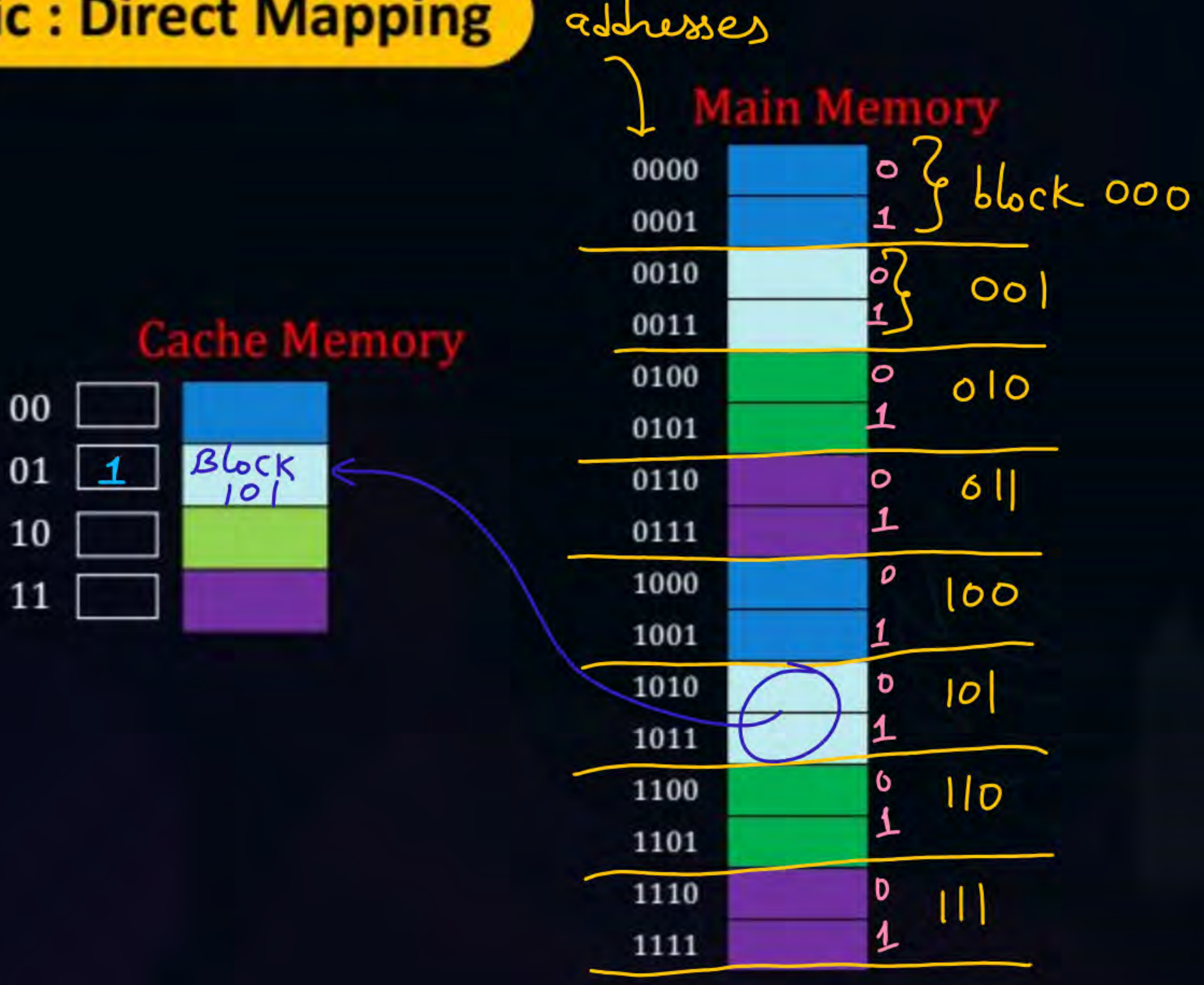
mm block no.

Tag	cm block no.

Topic : Direct Mapping

- Blocks in cache = 4 (00-11)
- Blocks in Main memory = 8 (000-111)
- Block Size = 2 Bytes
- Size of Cache memory = $4 * 2B = 8$ Bytes
- Size of Main memory = $8 * 2B = 16$ Bytes $= 2^4 B$
- Size of Main memory address = 4 bits
  (byte addressable)

Topic : Direct Mapping



mm address

4 bits

mm block no.	byte no. within block
3 bits	1 bit

or
(byte offset)

Tag	cm block no.	byte no.
1	2 bits	1 bit

Topic : Direct Mapping

CPU generates mm address $(1010)_2$

101	0
mm block	byte no.

bring mm block no. 101 at cache block 01 with tag 1

1	01	0
Tag	cm block no.	

↳ goto cm block no. 01 ⇒ no any block present there

⇑

miss

Topic : Direct Mapping

CPU generates mm address $(1011)_2$

101	1
mm block no.	byte no.

→ CPU accesses byte 1 of this block ← hit

1	01	1
Tag	cm block no.	

goto cm block no. = 01 ⇒ Tag in cache here = 1

Tag in CPU's request ⇒ 1

Topic : Direct Mapping

no. of bits needed for byte offset = $\log_2$(block size)

no. of blocks in cache = $\dfrac{\text{cache size}}{\text{block size}}$

no. of bits needed for cm block no. = $\log_2$(no. of blocks in cache)

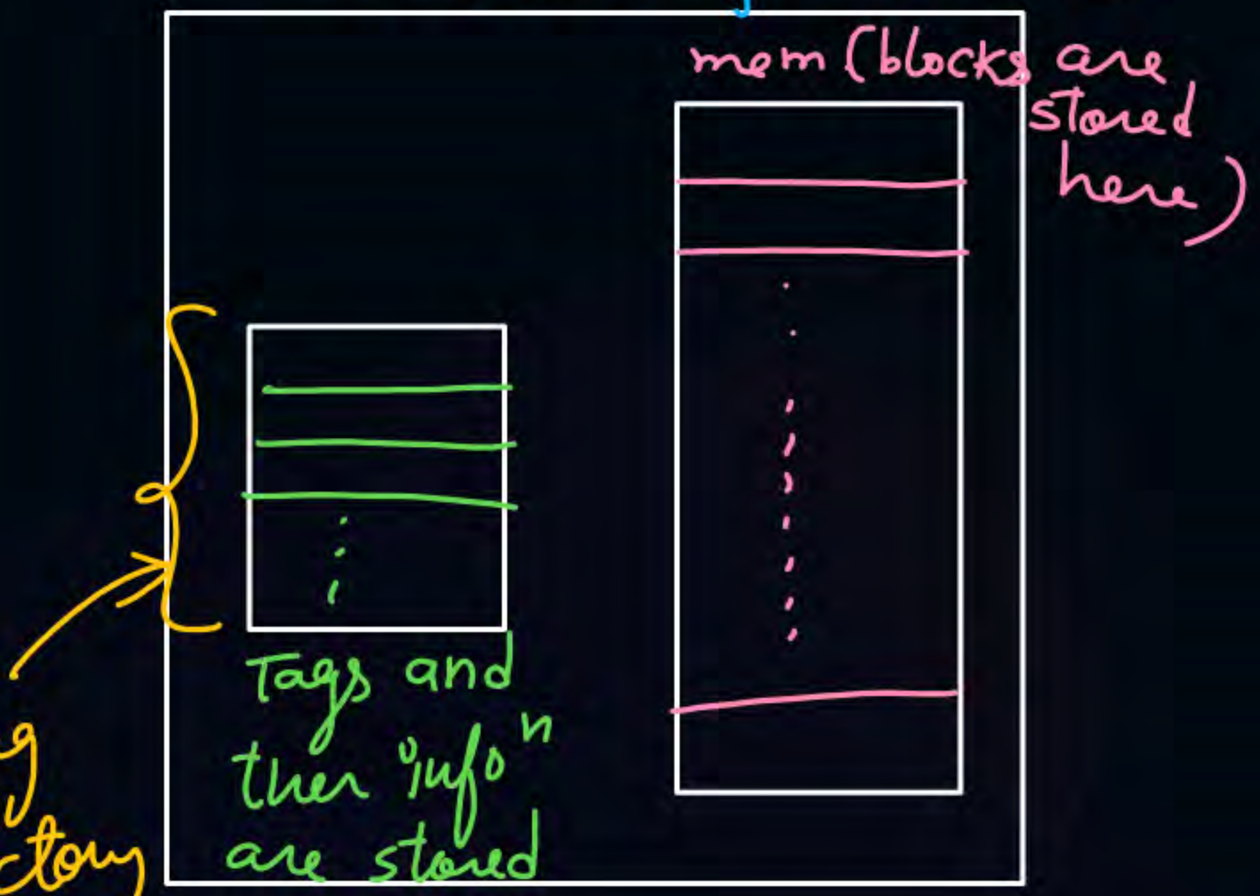
Topic : Indexing in Direct Mapping

cm block no. is known as index in cache

(cm line no.)

Topic : Cache Controller

Cache chip hardware



Tag directory or metadata

Tag directory size

= no. of blocks in cache * Tag bits

[NAT]

$2^3 B \Rightarrow$ byte offset = 3 bits

#Q. Consider a direct mapped cache of size 512B with block size 8 Bytes. The CPU generates 13-bits addresses.

1. Number of bits for byte offset?
2. Number of blocks in cache?
3. The number of bits needed for cache indexing?
4. The number of tag bits?
5. Tag Directory size?

Happy Learning

THANK - YOU